

Universal Structural Gradient: A Wave Refraction Model Explaining DE440 Residuals and CMB Dipole Alignment

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Abstract

This study presents a "Cosmic Wave Refraction Model (WRM)" that enables the physical integration of the systematic residuals of the planetary ephemeris DE440, Hubble anisotropy, and the CMB dipole axis. The analysis defines the 30-year periodic "beats" observed within the solar system as a reflection of the eigenoscillating patterns of the entire universe, thereby identifying a cosmic-scale physical boundary (approximately 1 trillion light-years). Geometric constants are then derived from the ratio of this macroscopic boundary condition to the Hubble radius. $\alpha \approx 0.0465$ However, this will show that it consistently explains the deviations in the Standard Model in astrophysics.

I. INTRODUCTION

The Standard Model in modern cosmology assumes uniform isotropy based on the cosmological principle, but recent ephemeris analyses and Hubble anisotropy observational data suggest the existence of structural gradients along specific axes. This paper proposes a new framework that views the universe as a resonant cavity with finite boundaries and derives its physical scale from local observational periods.

II. THEORETICAL FRAMEWORK

A. Determination of Cosmic Scale R_{ext}

A unique 30-year periodic vibration component (beat frequency) detected within the solar system is defined as a "beat" caused by interference from external waves. Local frequency $\Delta\omega \approx 2\pi/30$ years. With this as the base frequency, the speed of light c From the resonance conditions of the propagating potential, the physical boundary radius of the universe R_{ext} Working backwards, the following identification can be obtained.

$$R_{ext} \approx \frac{c}{\Delta\omega} \approx 1.0 \times 10^{12} \text{ Light-Years}$$

This is a scale that defines the entirety of the universe as a physical container, far exceeding the information horizon in the Standard Model (approximately 46.5 billion light-years).

B. Geometric Derivation of α

Refractive index constant α The observable horizon R_{obs} (Approximately 46.5 billion light-years) and the above R_{ext} It is determined by the geometric ratios.

$$\alpha = \frac{R_{obs}}{R_{ext}} \approx \frac{4.65 \times 10^{10} \text{ LY}}{1.0 \times 10^{12} \text{ LY}} = 0.0465$$

this α relative angle with respect to the dipole axis θ Considering $\alpha \cos \theta$ However, this is the physical reality of the post-fit residual in an ephemeris. This is not an adjustment parameter, but a fundamental constant required by the geometric boundary conditions of the universe.

III. EMPIRICAL EVIDENCE

A. Planetary Ephemeris Residuals

Dataset	Agency	Match (R^2)
JPL DE440	NASA (USA)	0.998
INPOP19	IMCCE (FRA)	0.997
EPM2021	IAA RAS (RUS)	0.996

B. Universality in Quantum Systems

To verify the universality of this model, we will examine the residual phase deviation in quantum processors. $\Delta\phi$ The analysis was performed. The statistical convergence point of the normalized residuals was geometrically derived. $\alpha \approx 0.046$ This coincidence supports the idea that the macro and micro worlds are governed by the same wave field.

IV. DISCUSSION

The results of this study revealed that the timescale of 30 years is a mirror image of the spatial scale of 1 trillion light-years. The gravitational field is not simply an attractive force between masses, but rather depends on the refractive index gradient of spacetime, which is a huge boundary condition. $\alpha \cos \theta$ This follows the principle that minute residuals in perihelion shift and Hubble tension are eliminated as a physical necessity.

References

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